Name :- Aakanksha . K. Bahde Std :- 3rd year (5th sem)
Sub :- Chemistry Assignment
Topic :- Photochemistry

Q.1. give statement of lambert's law and derive it's equation.

⟹

- Statement :- In 1759, J.H Lambert derived a relationship between the amount of light absorbed and the thickness or depth of the absorbing material. According to this law, when a beam of monochromatic light passes through a homogeneous absorbing medium, the rate of decrease of the intensity of the radiation with thickness of absorbing medium is proportional to the intensity of the incident radiation.
- equation:

Mathemically this law can be expressed as

$$-\frac{dI}{dx} \propto I$$

$$-\frac{dI}{dx} = kI$$

where, I = Intensity of incident light.

x = Thickness of the medium,

k = constant called absorption coefficient.

on rearranging Equation (1) we get,

$$\frac{dI}{I} = -Kdx$$

Integrating Equation (2) between the limits.

$I = I_0$ at $x = 0$ and $I = I$ at $x = x$, we get

$$\int_{I_0}^{I} \frac{dI}{I} = -K \int_{x=0}^{x=x} dx$$

or $$\ln \frac{I}{I_0} = -Kx$$

$$\frac{I}{I_0} = e^{-Kx}$$

$$I = I_0 e^{-Kx}$$

According to equation (5), the intensity of beam of monochromatic radiation decreases exponentially with increase in the thickness 'x' of the absorbing substance. ~~The~~ This is Lamberts law. Equation (4) can be written as;

$$2.303 \log \frac{I}{I_0} = -Kx \quad \text{———} (7)$$

$$\log \frac{I}{I_0} = \frac{-K}{2.303} x \quad \text{--------- (8)}$$

$$\log \frac{I}{I_0} = \varepsilon x \quad \text{--------- (9)}$$

$$\log \frac{I_0}{I} = -\varepsilon x \quad \text{--------- (10)}$$

where, $\varepsilon = \frac{-K}{2.303}$ and is called extinction coefficient. It depends upon nature of the absorbing substance and wavelength of incident light.

Q.2. Explain phenomena of fluorescence and phosphorescence with the help of Jablonski diagram?

⇒ The absorption of light may result in a number of phenomena for instance, the light absorbed may be re-emitted almost instantaneously (within 10^{-8} second) in one or more steps. This phenomenon is known as fluorescence. The emission in fluorescence ceases with the removal of the source of light. sometimes the light absorbed is given out slowly and even long after the removal of the source of light. This phenomenon is known as phosphorescence.

The phenomena of fluorescence and phosphorescence are best explained with the help of Jablonski diagram.

In order to understand this diagram, we need to define some terminology. Most molecules have an even number of electrons thus in the ground state, all electrons are spin paired. The quantity 2s+1, is known as the spin multiplicity of a state and s is the total spin.

When spins are paired (↓↑) as shown in fig. 1 (a) the upward orientation of the electron spin is cancelled by the downward orientation so that $s = 0$. This is illustrated below $s_1 = 1/2$ & $s_2 = -1/2$ so that $s = s_1 + s_2 = 1/2 - 1/2 = 0$.

Hence, $2s + 1 = 1$. Thus the spin multiplicity of molecule is 1. we express it by saying that the molecule is in the singlet ground state.

When upon the absorption of a photon of suitable energy $h\nu$, one of the unpaired electrons goes to a higher energy level (excited state). the spin orientation of two single electrons may be either parallel ↑↑ or antiparallel, ↓↑ as shown in fig 3 (b) and 3 (c) respectivally. If the spins are parallel, as in fig. 3 (b), then

$s = s_1 + s_2 = 1/2 + 1/2 = 1$ so that $2s + 1 = 3$

Thus, the spin multiplicity of the molecule is 3. This is expressed by saying that the molecule in the triplet excited state.

(b) (a) (c)

fig. 3 spin orientation on the absorption of a light photon

If, however, the spins are antiparallel as shown in Fig 3(c),
$S = s_1 + s_2 = \frac{1}{2} - \frac{1}{2} = 0$ so that $2S+1 = 1$.
Thus, the spin multiplicity of the molecule is 1. This is expressed by saying that the molecule is in the singlet excited state.

Since the electron can jump to any of higher electronic state depending upon the energy of the photon absorbed, we get a series of singlet excited states 'S_n' where $n = 1, 2, 3, 4$ ...... and a series of triplet excited states, 'T_n' where $n = 1, 2, 3, 4$ ...

Thus S_1, S_2, S_3 etc. are known as the first singlet excited state, second singlet excited state, third singlet excited state etc. Similarly T_1, T_2, T_3 etc. are called the first triplet excited state, second triplet excited state, third triplet excited state etc.

It has been shown quantum mechanically that a singlet excited state has higher energy than the corresponding triplet excited state. Accordingly the energy sequence is as follows:-

$ES_1 > ET_1$; $ES_2 > ET_2$; $ES_3 > ET_3$ and so on

On absorption of light photon, the electron of the absorbing molecule may jump from S_0 to S_1, S_2 or S_3 singlet excited state depending upon the energy of the light photon absorbed, as shown in Jablonski diagram 4.

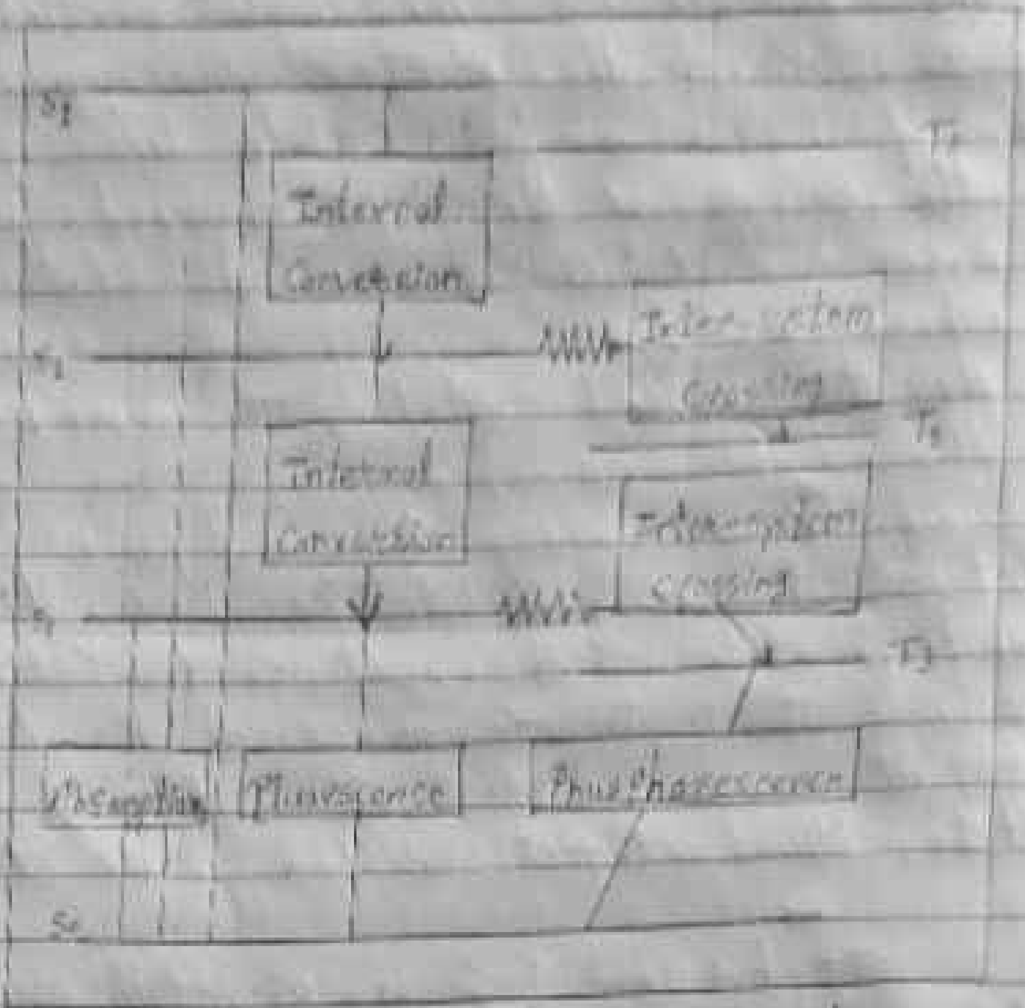

Fig. 4. The Jablonski diagram depicting various photophysical processes.

For each singlet excited state (S_1, S_2, S_3 etc.) there is corresponding triplet excited state, is said to be activated. Thus $A + h\nu \rightarrow A^*$
where A is the molecule in the ground state and A^* is the molecule in the excited state.

Q.5. Explain kinetics of photochemical decomposition of HI ?

⟹ The decomposition of HI takes place by the radiation in the region 2000-2200 Å. The following mechanism has been suggested for the photochemical decomposition of HI.

i) $HI + h\nu \xrightarrow{k_1} H + I$

Rate = $k_1 I_a$.

ii) $H + HI \xrightarrow{k_2} H_2 + I$

Rate - $k_2 [H][HI]$

iii) $I + I \xrightarrow{k_3} I_2$

Rate - $k_3 [I]^2$

Net reaction is $2HI \longrightarrow H_2 + I_2$

The rate law for decomposition of HI is given by

$$-\frac{d[HI]}{dt} = k_1 I_a + k_2 [HI][H] \quad \text{(I)}$$

Since hydrogen atoms are shorter lived, applying steady state treatment, we get

$$\frac{d[H]}{dt} = k_1 I_a = k_2 [H][HI]$$

$$\therefore [H] = \frac{k_1 I_a}{k_2 [HI]}$$

Substituting this value of [H] in eq. 1 we get
~~Rat~~
Rate of formation the rate of decomposition of HI

$$k_1 I_a = k_2 [H][HI]$$

$$= k_1 I_a + k_2 [HI] \frac{k_1 I_a}{k_2 [HI]}$$

$$= \frac{-d[HI]}{dt} = k_1 I_a + k_1 I_a$$

$$= \cancel{9}\, 2 k_1 I_a$$

Thus the rate of dissociation of HI depends upon the intensity of the incident radiation.

Q. Difference between thermal & photochemical reactions:-

Thermal reaction	Photochemical reaction
1. In thermal reaction the energy of activation is provided by the collisions.	In photochemical reaction the energy required is gained through the absorption of quanta of visible or ultraviolet radiation.
2. Thermal reaction proceed with a decrease in free energy.	photochemical reaction proceed with a increase in free energy.
3. Rate of thermal reaction ~~are~~ ~~fast~~ depends upon temperature.	Rate of photochemical reaction is independent of temperature but depends on the intensity of radiation used.
4. thermal reaction are fast	photochemical reaction ~~is~~ are slow.
5. Example of thermal reaction	Example of photochemical reaction
i) $N_2 + 3H_2 \longrightarrow 2NH_3$	i) $2HBr \xrightarrow{light} H_2 + Br_2$
ii) $H_2 + I_2 \longrightarrow 2HI$	ii) $H_2 + Cl_2 \xrightarrow{light} 2HCl$

Q.5. light of ~~mat~~ intensity $3.2 \times 10^{-?}$ einstein/sec fall on the surface of a crystal where absorption coefficient is 3×10^{-6} cm^{-1} at the wavelength of light calculate the intensity of light at a depth of $100 A^{\circ}$ below the surface of the ~~cr~~ crystal.

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$I_0 = 3.2 \times 10^{-6}$ einstein/sec $\xrightarrow{I_0}$ Crystal $\xrightarrow{I}$ = ?

Here $x = 100 A^{\circ} = 10^{-6} cm$

$(10^{8} A^{\circ} = 1 cm)$

Absorption coefficient $= 3 \times 10^{-6} cm^{-1}$
(K)

$$\log \frac{I_0}{I} = \frac{K\ x}{2.303}$$

$$\log \frac{3.2 \times 10^{-6}}{I} = \frac{3 \times 10^{-6}}{2.303} \times 10^{-6}$$

$I = 4.58 \times 10^{7}$ einstein/sec